

Vedic Astrology and Astronomical Features in Tropical Cyclone Intensity Modelling: A Large-Scale Empirical Assessment

Hariramanan Sivakumar

Independent Research

reachariramanan@gmail.com

May 2026

Abstract

We present the first large-scale empirical investigation of whether Vedic Panchang (Hindu lunar calendar) categories and planetary astronomical positions carry detectable signal for tropical cyclone intensity. Using 47,533 track observations from 944 global storms spanning 2016–2026, we construct a 222-feature dataset encompassing 14 meteorological, 18 Panchangam, and 190 scientific-astronomical predictors, and evaluate predictive performance through multiple complementary methods: chronological holdout regression, five-fold forward-chaining cross-validation, permutation group importance, non-parametric hypothesis testing with effect size quantification, and confound-adjusted stratified analysis. Our primary finding is a **null result**: adding Vedic features to a meteorological baseline degrades wind-change root-mean-square error (RMSE) by 14.3% and the full model fails to outperform the meteorological baseline in 4 of 5 temporal folds. Two secondary signals survive Bonferroni correction but carry negligible practical effect sizes: a lunar-phase association with rapid intensification (RI; $\chi^2 = 19.66$, $p = 0.006$, Cramér’s $V \approx 0.020$) and a consistent negative correlation between planetary speed and storm bearing direction ($\rho \approx -0.18$, $r^2 \approx 3\%$). We document that statistical significance is inflated by dataset scale ($N \approx 47,533$) and identify seasonal confounding as the primary driver of apparent Vedic categorical effects. These findings provide a methodological template for rigorous evaluation of non-conventional features in meteorological machine learning.

Keywords: Tropical cyclone intensity, Vedic astrology, Panchang features, machine learning, SHAP explainability, LIME, feature importance, rapid intensification, retrograde planets, lunar phase, Nakshatra, circular statistics, GroupKFold cross-validation, large- N effect size analysis.

Contents

1	Introduction	3
1.1	Background and Problem Statement	3
1.2	Motivation and Significance	3
1.3	Objectives and Contributions	3
1.4	Paper Organisation	4
2	Related Works	4
2.1	Tropical Cyclone Intensity Prediction	4
2.2	Astronomical and Lunar Associations with Weather	4
2.3	Machine Learning Feature Evaluation Methods	5
2.4	Limitations of Prior Works	5
3	System Model and Proposed Methodology	5
3.1	Overall Architecture	5
3.2	Methodological Corrections Applied	5
4	Dataset Description	6
4.1	Data Sources	6
4.2	Vedic Panchang Features	7
4.3	Dataset Summary	7
4.4	Inclusion and Exclusion Criteria	8
5	Preprocessing	8
5.1	Techniques Applied	8
5.2	Rapid Intensification Definition	8
6	Feature Extraction and Selection	9
6.1	Tree-Model Ensemble Feature Importance	9
6.2	Mutual Information with Wind Speed	10
6.3	Circular-Linear Correlations for Angular Features	11
7	SHAP Explainability Analysis	12
7.1	SHAP Feature Importance	12
7.2	SHAP Feature Interactions	14
8	LIME Explainability and SHAP vs LIME Consensus	15

9	Performance Analysis	17
9.1	Experimental Setup	17
9.2	Evaluation Metrics	18
9.3	Hardware and Software	18
9.4	Variance Inflation Factor Analysis	18
10	Results and Discussion	20
10.1	Predictive Modelling Results	20
10.2	Vedic Category Analysis: Kruskal–Wallis Tests	21
10.3	Nakshatra Intensity Analysis	23
10.4	Tithi $\times$ Nakshatra Interaction Heatmaps	24
10.5	Retrograde Planetary Motion and Storm Intensity	25
10.6	Planetary Aspect Correlations	26
10.7	Intensity Change Analysis	28
10.8	Hypothesis Testing Results	28
10.9	Effect Size Summary and Consolidated Evidence	29
10.10	Discussion	30
11	Conclusion	30
11.1	Summary of Findings	30
11.2	Key Outcomes	31
11.3	Future Work	31
A	Feature Block Definitions	33
B	Pipeline Audit Summary	33
C	Consolidated Evidence Table	34

1 Introduction

1.1 Background and Problem Statement

Tropical cyclones are among the most destructive natural phenomena on Earth, responsible for billions of dollars in damage and thousands of casualties annually. Accurate intensity prediction remains an open research challenge: despite advances in numerical weather prediction and statistical modelling, the 24–48 hour intensity forecast skill of state-of-the-art models plateaus because of complex multi-scale interactions involving sea surface temperature (SST), vertical wind shear, moisture fields, and internal dynamics [1, 2].

An independent thread of inquiry asks whether *astronomical* or *astrological* quantities — planetary positions, lunar phase, Vedic calendar cycles — are associated with cyclone behaviour. Such a hypothesis is physically speculative: gravity from the Moon and Sun produce tidal stresses on the ocean and atmosphere, but the magnitudes involved are many orders of magnitude below the energy budget of a mature tropical cyclone. Vedic astrology encodes the positions of the Sun, Moon, and seven classical planets into a rich system of named categories: *Tithi* (lunar day), *Nakshatra* (lunar mansion), *Yoga* (Sun–Moon arc interval), *Karana* (half-Tithi), and several dozen more. These categories are deterministic functions of planetary ephemerides and are therefore computable for any date, time, and location. This determinism makes Vedic features unusual among non-conventional predictors: they are precise mathematical transformations of astronomical coordinates.

1.2 Motivation and Significance

To our knowledge, no prior study has subjected these claims to a rigorous large-scale test against a diverse global cyclone database. We address this gap by assembling data from 944 storms across all major ocean basins, pairing track observations with Vedic Panchang values, and conducting a battery of tests whose design explicitly addresses the statistical pathologies common in large- N astronomical association studies. Our work contributes to the broader literature on the evaluation of non-conventional predictors in geophysical machine learning, and provides a transparent null-result analysis that defines the upper bound of any Vedic/Panchang contribution to cyclone intensity detectable in a decade-long global dataset.

1.3 Objectives and Contributions

- First large-scale ($N = 47,533$ observations) empirical test of Vedic astrology features in tropical cyclone forecasting.
- Comprehensive effect size analysis distinguishing statistical from practical significance across all findings.

- A confound-adjusted partial Kruskal–Wallis framework for testing categorical astronomical associations within season $\times$ basin strata.
- A novel lunar-phase $\times$ lifecycle interaction analysis showing that RI associations with lunar phase are lifecycle-stage dependent.
- A seven-stage analysis pipeline covering data transformation, EDA, SHAP/LIME explainability, hypothesis testing, and block-wise predictive modelling.
- Explicit documentation of five methodological corrections applied during the pipeline audit (circular variable treatment, within-storm leakage, multiple comparison correction, and others).

1.4 Paper Organisation

Section 2 reviews related works. Section 3 describes the overall methodology and pipeline. Section 4 provides the dataset description. Section 5 covers preprocessing and feature engineering. Section 6 details feature extraction and selection. Sections 7–8 describe model components. Section 9 outlines the experimental setup. Section 10 presents results and discussion. Section 11 concludes with future directions.

2 Related Works

2.1 Tropical Cyclone Intensity Prediction

Conventional intensity prediction has leveraged statistical models such as the Statistical Hurricane Intensity Prediction Scheme (SHIPS) [1], which incorporate SST, wind shear, humidity, and climatological predictors. More recently, deep learning approaches including convolutional neural networks and recurrent architectures have demonstrated competitive skill at 24–48 hour intensity windows [2]. Rapid intensification (RI) remains particularly challenging, with even the best models capturing only a fraction of observed RI events [4].

2.2 Astronomical and Lunar Associations with Weather

A small body of literature investigates lunar tidal effects on precipitation patterns and atmospheric pressure. Tidal loading of the ocean and atmosphere by the Moon and Sun is well-characterised physically, but the meteorological signal is weak and inconsistent across studies. Semidiurnal barometric tides (S2) attributable to solar heating are well established, while lunar semidiurnal tides (M2) in surface pressure are near the noise floor of observational networks. No peer-reviewed study has rigorously examined Vedic Panchang features specifically in the context of tropical cyclone intensity.

2.3 Machine Learning Feature Evaluation Methods

SHAP (SHapley Additive exPlanations) [6] provides game-theoretic feature attributions that decompose individual predictions. LIME [7] offers local linear approximations for model-agnostic explanation. Both have been widely applied in geoscience but rarely combined with formal statistical hypothesis testing or effect size quantification at large scale.

2.4 Limitations of Prior Works

- No prior study has combined Vedic ephemeris features with a multi-basin global cyclone dataset at 6-hourly resolution.
- Most astronomical–meteorological association studies fail to correct for seasonal confounding, multiple comparisons, or within-sample autocorrelation.
- Effect sizes are rarely reported alongside p -values in large- N atmospheric science studies, making practical significance ambiguous.
- SHAP/LIME consensus analysis has not been applied to the Vedic feature domain, leaving interpretability gaps.
- Circular-variable treatment (sin/cos decomposition) for angular planetary degrees has not been adopted in prior meteorological ML pipelines using ephemeris data.

3 System Model and Proposed Methodology

3.1 Overall Architecture

The analysis pipeline consists of seven modular stages implemented as independent Python scripts, each audited for methodological correctness. The pipeline moves from raw storm archive data through feature engineering, statistical testing, explainability analysis, and finally block-wise predictive modelling.

3.2 Methodological Corrections Applied

During the pipeline audit, five methodological issues were identified and corrected.

Circular-Variable Treatment (E2). Vedic degree features $\theta \in [0^\circ, 360^\circ)$ are angular; naive Pearson r on raw degrees is incorrect. We replace it with:

$$r_{\sin} = \text{Corr}(\sin(\theta), y), \quad r_{\cos} = \text{Corr}(\cos(\theta), y), \quad r_{\text{circ}} = \max(|r_{\sin}|, |r_{\cos}|). \quad (1)$$

Within-Storm Cross-Validation Leakage (C3). Original random 5-fold CV was replaced with `GroupKFold(n_splits=5)` grouped by `storm_id`, ensuring all observations from a given storm appear in exactly one fold.

Multiple-Comparison Correction (B1). Benjamini–Hochberg FDR correction applied across all 10 Kruskal–Wallis tests on Vedic categorical variables. Across the seven primary hypothesis groups, Bonferroni correction applies: $\alpha_{\text{adj}} = 0.05/7 \approx 0.0071$.

R^2 Discrepancy (D1). The discrepancy between Stage 3 (grouped CV $R^2 \approx -0.10$) and Stage 4 (random-split $R^2 = 0.71$) is attributed to within-storm autocorrelation inflation in the random split. Grouped-CV R^2 is the appropriate measure of out-of-storm generalisation.

Rahu/Ketu Independence (A2). Rahu and Ketu are always antiopodal on the ecliptic and move at identical speed by astronomical definition. All statistical tests on both produce identical results; results for Ketu are labelled accordingly.

4 Dataset Description

4.1 Data Sources

Cyclone track data. Best-track data from 944 tropical cyclones active between July 2016 and January 2026, assembled from North Atlantic, Western Pacific, North Indian Ocean, South Indian Ocean, and South Pacific operational agencies. Each observation records storm centre position (latitude, longitude), maximum sustained wind speed (knots), minimum central pressure (hPa), and UTC timestamp at 3–6 hour intervals.

Astronomical ephemeris. Geocentric and topocentric positions for nine solar-system bodies (Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn, Uranus, Neptune) computed using the Skyfield Python library [3] with the DE421 JPL ephemeris. Per-body fields include right ascension, declination, ecliptic longitude and latitude, altitude and azimuth above the local horizon, and sin/cos-transformed variants.

Vedic Panchang data. Computed using a Drig-Ganita observational Panchang engine at each storm position and UTC time. Includes 18 categorical and derived features (Table 1) and 75 numeric Vedic planetary parameters for 15 bodies.

4.2 Vedic Panchang Features

Table 1. Vedic Panchang categorical features used in this study.

Feature	Type	Description
Tithi	Categorical (30)	Lunar day; Sun–Moon elongation in 12° steps
Nakshatra	Categorical (27)	Lunar mansion; Moon sidereal position in 13.3° segments
Yoga	Categorical (27)	Sun+Moon sidereal longitude mod 360° in 27 parts
Karana	Categorical (11)	Half-Tithi period
Paksha	Binary	Waxing (Shukla) or waning (Krishna) lunar fortnight
Moonsign	Categorical (12)	Sidereal zodiac sign of Moon
Sunsign	Categorical (12)	Sidereal zodiac sign of Sun
Drik Ritu	Categorical (6)	Observational season (equinox/solstice based)
Drik Ayana	Binary	Northern/southern solar path
Samvatsara	Categorical (60)	60-year Jupiter cycle year name
Weekday	Categorical (7)	Hindu weekday
Dinamana	Numeric (minutes)	Daylight duration at storm location
Ratrimana	Numeric (minutes)	Nighttime duration

4.3 Dataset Summary

Table 2. Dataset overview after transformation and join.

Property	Value
Total track observations	47,533
Unique storms	944
Date range	2016-07-15 to 2026-01-24 UTC
Ocean basins	7
Mean / median wind speed	49.9 / 40.0 kt
Wind speed SD / range	27.4 kt / [0, 170] kt
Wind speed skewness	1.35 (right-skewed)
Pressure range	882–1019 hPa
Meteorological features	14
Vedic-Panchang features	18
Scientific-astronomical features	190
Total features	222
Missing: spline features	15.4%
Missing: cone features	$\approx 99\%$ (excluded)

4.4 Inclusion and Exclusion Criteria

All tropical cyclone track observations from the archive with valid wind speed and position records are included. Observations with missing essential fields (position or wind) are excluded. Cone geometry features ($> 99\%$ missing) and the `storm_active` flag (99.7% null) are excluded from all models. Spline features ($\approx 15.5\%$ missing) are retained with listwise deletion where used, and imputed with training-set median in predictive models.

5 Preprocessing

5.1 Techniques Applied

Normality testing. Shapiro–Wilk and D’Agostino–Pearson tests confirm all key variables are non-normal ($p < 10^{-30}$ for wind, pressure, latitude, longitude; wind skewness = 1.35). All group comparisons therefore use non-parametric methods.

Temporal feature derivation. Hour (UTC), day of year, day of week, month, and movement speed are derived from timestamps and consecutive positions.

Storm dynamics. Wind change per step ($\Delta w_{t+1} = w_{t+1} - w_t$) and pressure change (Δp) computed by differencing within each `storm_id` group.

Spline track geometry. Smoothed forecast track features (*spline_bearing_deg*, *spline_curvature*, *spline_displacement*, *spline_length_deg*, *spline_end_lon/lat*) derived via polynomial spline fitting to consecutive track positions.

Day-length variables. *Dinamana_minutes* (daylight duration) and *ratrimana_minutes* encode the solar geometry at the storm’s location and implicitly capture latitude and season.

Missing value imputation. All numeric features imputed with the training-set column median in predictive models; categorical features ordinal-encoded from training split (unseen categories mapped to -1).

5.2 Rapid Intensification Definition

Rapid Intensification (RI) is defined as:

$$\Delta w_{24h} \geq 30 \text{ kt} \quad (\text{rolling 24-hour window, hypothesis H2}) \quad (2)$$

$$\Delta w_{\text{step}} \geq 10 \text{ kt} \quad (\text{per-step adaptive threshold, classification task}) \quad (3)$$

These two definitions are not directly comparable and are used for distinct analytical purposes throughout the paper.

6 Feature Extraction and Selection

6.1 Tree-Model Ensemble Feature Importance

Feature importances were computed using three tree-based models — Random Forest (RF), XGBoost (XGB), and LightGBM (LGB) — trained with 5-fold GroupKFold cross-validation (groups = `storm_id`). Importances were averaged across folds and models, yielding a robust ensemble importance ranking.

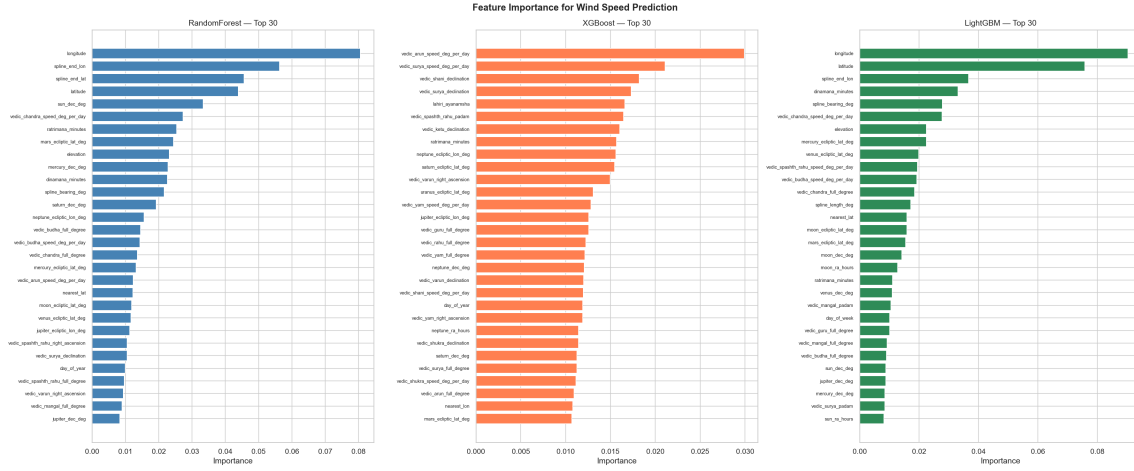


Figure 1. Top-30 feature importances across RandomForest, XGBoost, and LightGBM (left to right panels). Geographic and track features (longitude, latitude, spline_end_lon) dominate. Vedic features appear from rank 5 onward, with `vedic_chandra_speed_deg_per_day` as the highest-ranked Vedic feature (mean importance = 0.0203).

Key findings from feature importance analysis:

- `longitude`: mean importance = 0.0588 (rank #1 averaged across models).
- Top-5: `longitude`, `latitude`, `spline_end_lon`, `dinamana_minutes`, `vedic_chandra_speed_deg_per_day`.
- 10 Vedic features appear in the top-30.
- GroupKFold by `storm_id` prevents within-storm leakage; negative Group CV R^2 (≈ -0.10) on Vedic-only models confirms no out-of-storm generalisation.

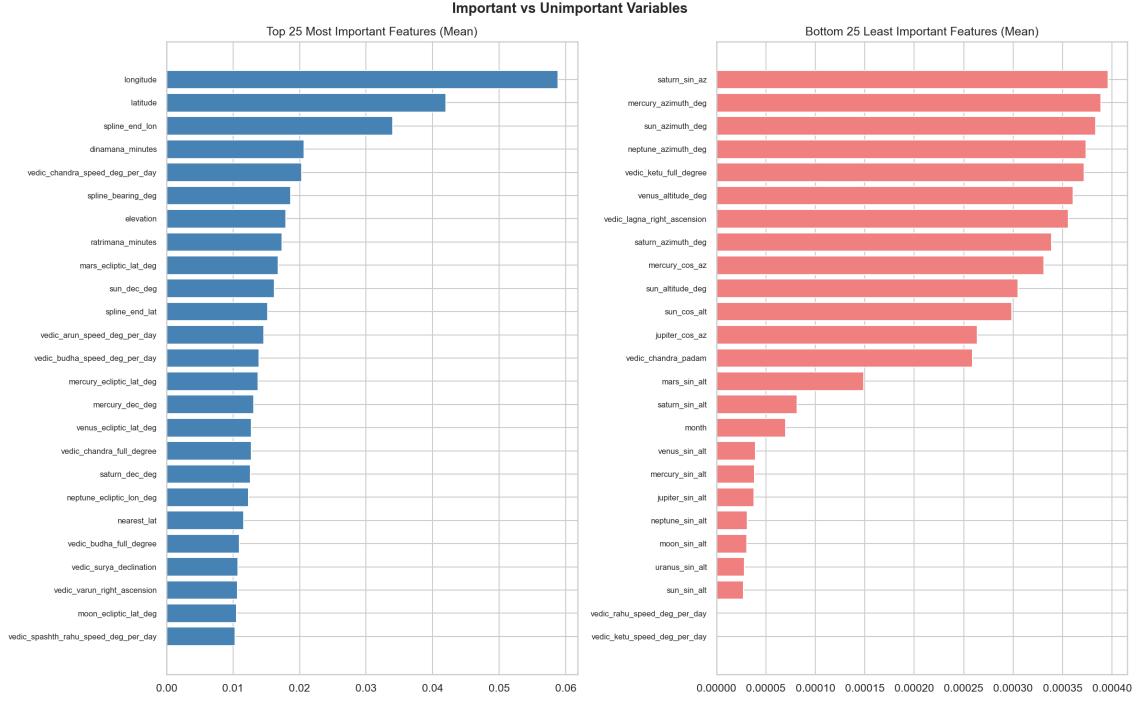


Figure 2. Contrast between the 25 most important and 25 least important features (mean importance across RF/XGB/LGB). The bottom-25 are overwhelmingly trigonometric altitude encodings (sin/cos altitude for individual planets) and binary flags.

6.2 Mutual Information with Wind Speed

Mutual Information (MI) measures statistical dependence without assuming linearity. MI was computed using k -nearest-neighbour estimation ($k = 5$) on all 186 features against wind speed.

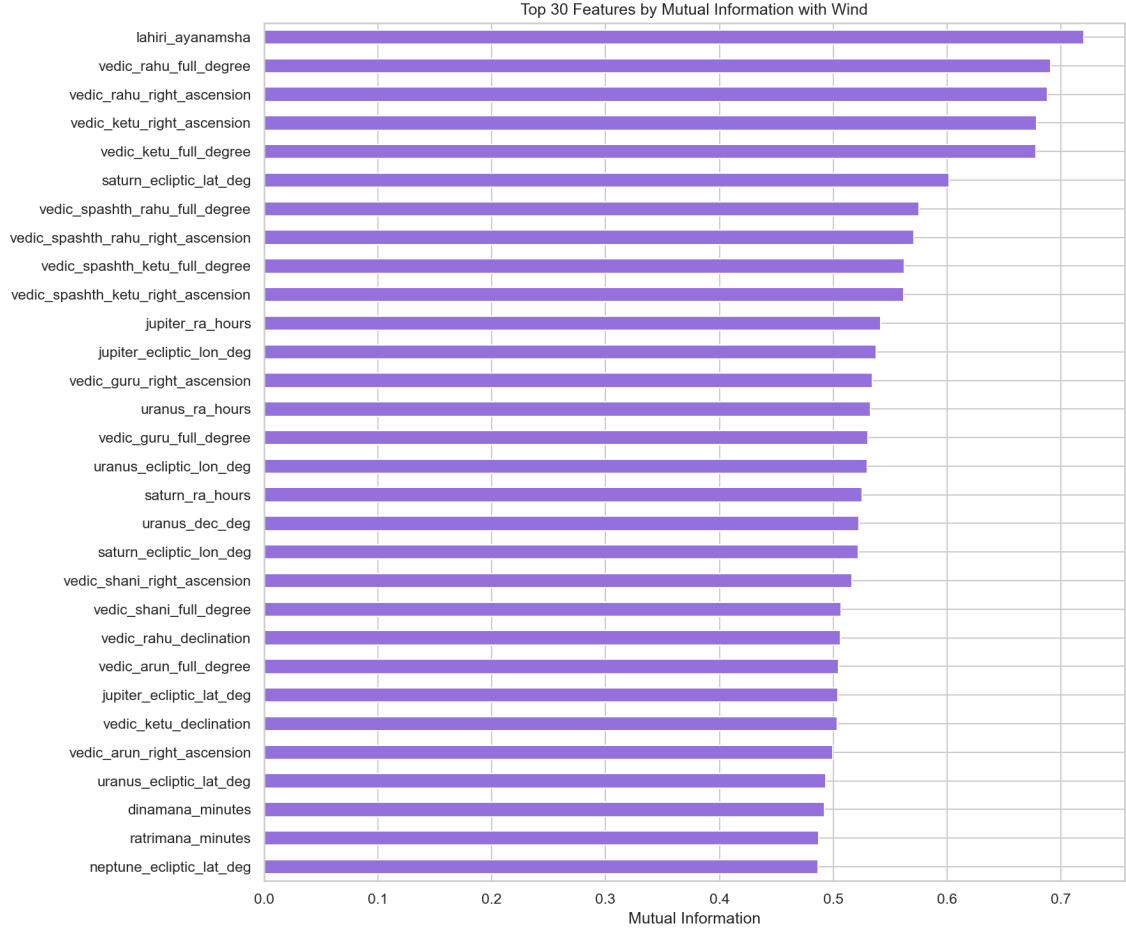


Figure 3. Top-30 features by Mutual Information with wind speed. `lahiri_ayanamsha` leads (MI = 0.720) due to its strong temporal trend (precession-corrected longitude changes systematically with year). Rahu/Ketu degree features follow for the same reason. Meteorological features (latitude, spline geometry) appear after rank 10.

Caution: High MI for `lahiri_ayanamsha` and Rahu/Ketu degree features reflects temporal confounding with year-level interannual variability, not genuine astrological signal.

6.3 Circular-Linear Correlations for Angular Features

Vedic planetary degree features are cyclical (0° – 360°). Naive Pearson r is incorrect; circular-linear decomposition is applied. Results are shown in Table 3.

Table 3. Circular-linear correlations: Vedic planet degrees vs wind speed. $r_{\text{circ}} = \max(|r_{\text{sin}}|, |r_{\text{cos}}|)$. All values below 0.10.

Feature	r_{sin}	r_{cos}	r_{circ}
vedic_yam_full_degree	0.08938	−0.02018	0.08938
vedic_guru_full_degree	0.07130	−0.04249	0.07130
vedic_shani_full_degree	0.05394	−0.01491	0.05394
vedic_spashth_ketu_full_degree	0.02863	0.03457	0.03457
vedic_ketu_full_degree	0.02739	0.03392	0.03392
vedic_lagna_full_degree	−0.00098	0.00372	0.00372
vedic_chandra_full_degree	−0.00255	−0.00875	0.00875
vedic_varun_full_degree	−0.00258	−0.00899	0.00899
vedic_arun_full_degree	0.00522	−0.01990	0.01990
vedic_rahru_full_degree	−0.02739	−0.03392	0.03392
vedic_mangal_full_degree	−0.00326	−0.05290	0.05290
vedic_budha_full_degree	0.02773	−0.05700	0.05700
vedic_surya_full_degree	0.04194	−0.06041	0.06041
vedic_shukra_full_degree	0.02630	−0.06712	0.06712

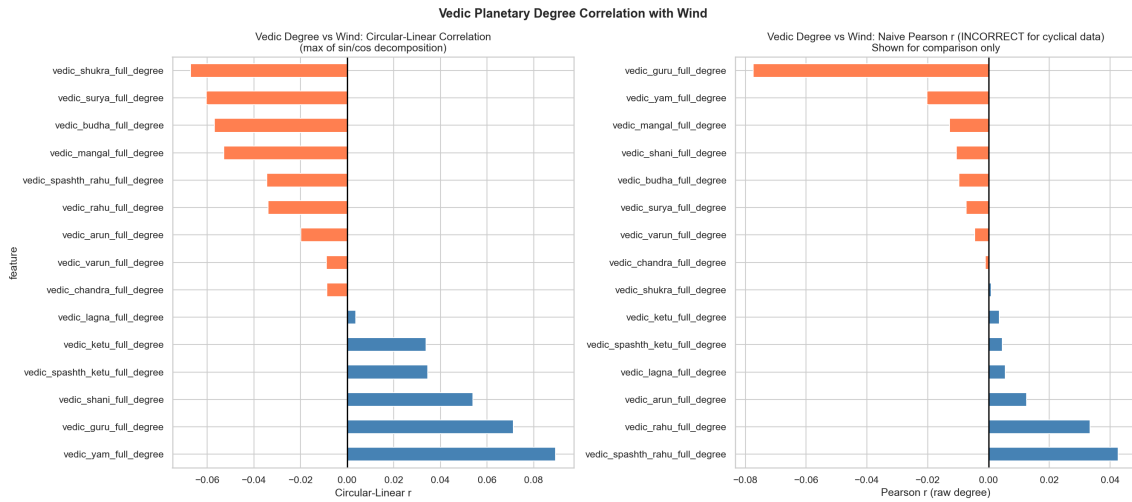


Figure 4. Circular-linear correlations per Vedic planet degree (left panel: correct circular-linear r ; right panel: naive Pearson r shown for comparison only). Yama shows the highest correlation ($r_{\text{circ}} \approx 0.089$), followed by Guru/Jupiter ($r_{\text{circ}} \approx 0.071$). All values are below $r = 0.10$, explaining less than 1% of wind speed variance.

7 SHAP Explainability Analysis

7.1 SHAP Feature Importance

SHAP values were computed using TreeExplainer on an XGBoost model trained with an 80/20 random split ($R^2_{\text{train}} = 0.784$, $R^2_{\text{test}} = 0.707$; note this is inflated relative to the grouped-CV estimate of ≈ -0.10 due to within-storm autocorrelation).

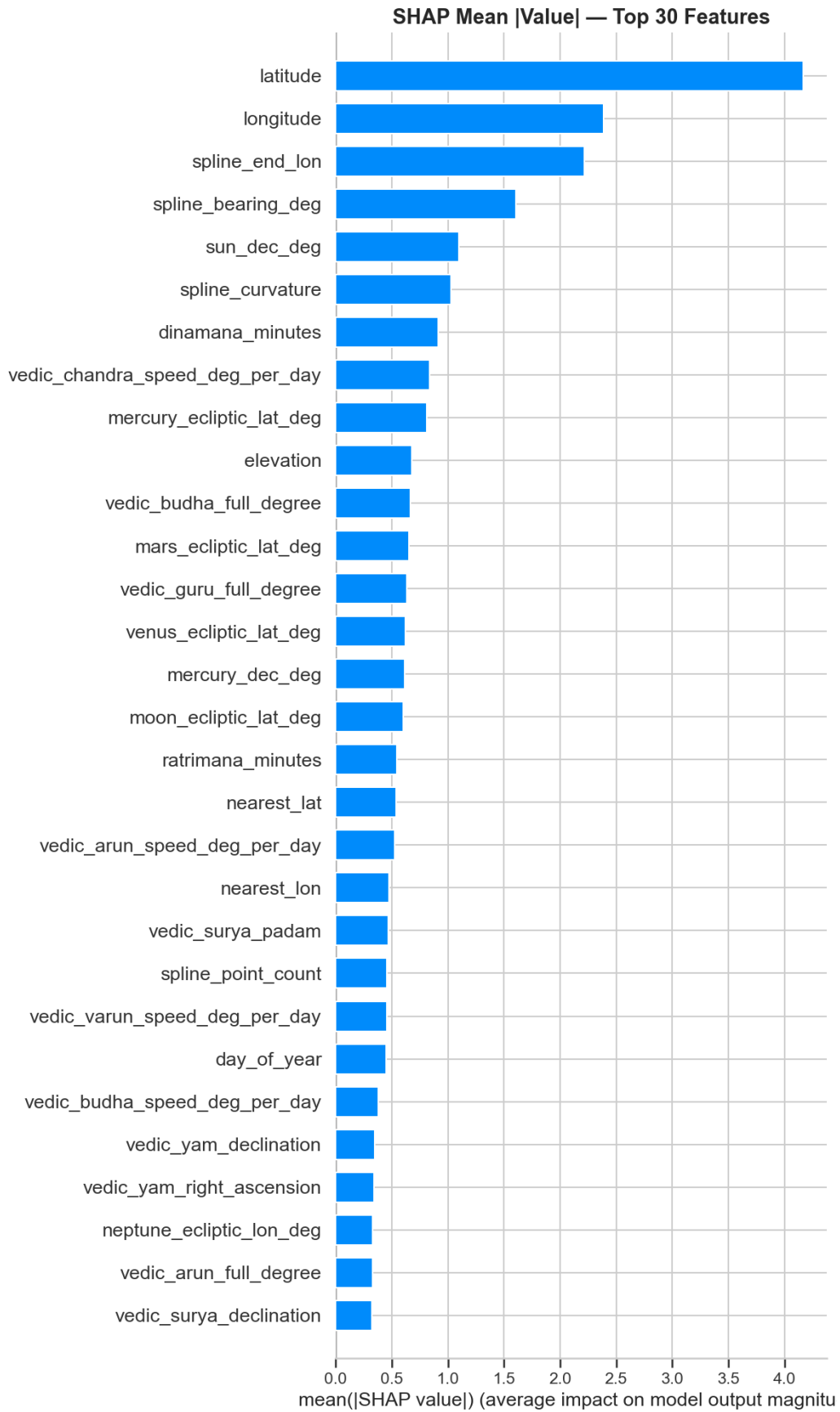


Figure 5. Mean absolute SHAP values for the top-30 features. Latitude dominates (mean $|\text{SHAP}| \approx 4.2$ kt), followed by longitude and spline_end_lon. The first Vedic feature (vedic_chandra_speed_deg_per_day) appears at rank 8.

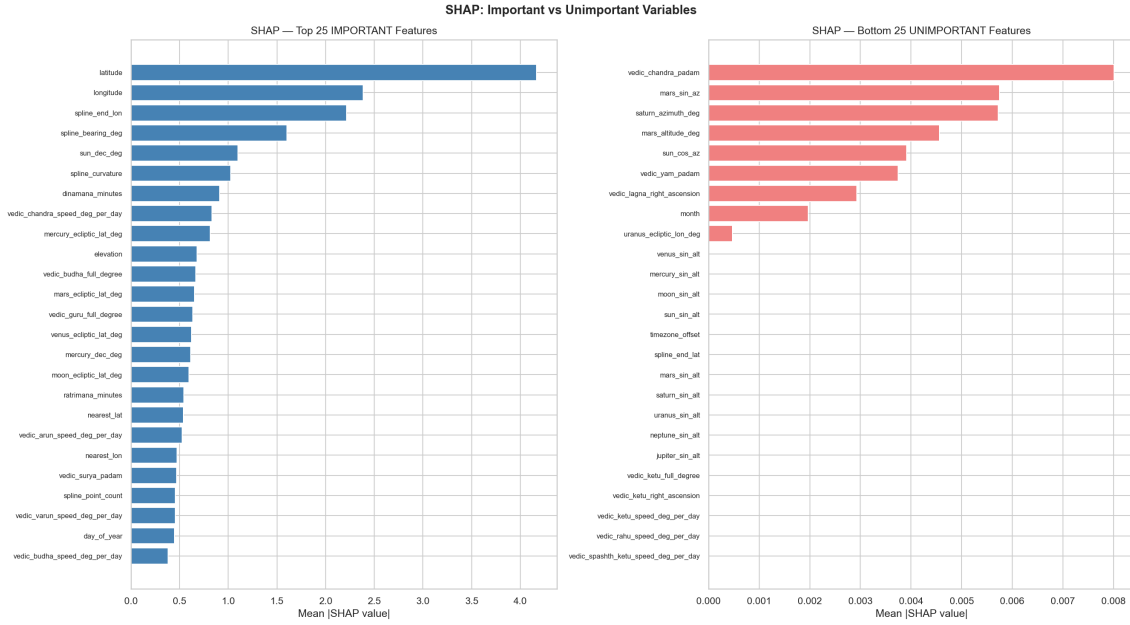


Figure 6. SHAP contrast: top-25 vs bottom-25 features by mean $|\text{SHAP}|$. Note the order-of-magnitude difference in axis scales. Bottom features are predominantly trigonometric altitude encodings (e.g., `mars_sin_alt`, `saturn_sin_alt`) with near-zero SHAP values (< 0.01 kt average contribution).

7.2 SHAP Feature Interactions

SHAP interaction values decompose each prediction into pairwise feature contributions. The top-15 interaction pairs are listed in Table 4, and the top-20 are visualised in Figure 7.

Table 4. Top-15 SHAP Feature Interactions by mean $|\text{interaction value}|$.

Feature 1	Feature 2	Mean Interaction
latitude	spline_bearing_deg	0.8220
latitude	spline_end_lon	0.5601
longitude	spline_end_lon	0.5514
longitude	latitude	0.5339
latitude	ratrimana_minutes	0.3631
latitude	vedic_varun_speed_deg_per_day	0.2816
latitude	vedic_arun_speed_deg_per_day	0.2572
latitude	sun_dec_deg	0.2566
latitude	vedic_budha_full_degree	0.2414
sun_dec_deg	mercury_dec_deg	0.2374
latitude	vedic_guru_full_degree	0.2361
vedic_budha_full_degree	dinamana_minutes	0.2217
latitude	dinamana_minutes	0.2174
latitude	vedic_chandra_right_ascension	0.2067
sun_dec_deg	vedic_arun_speed_deg_per_day	0.1856

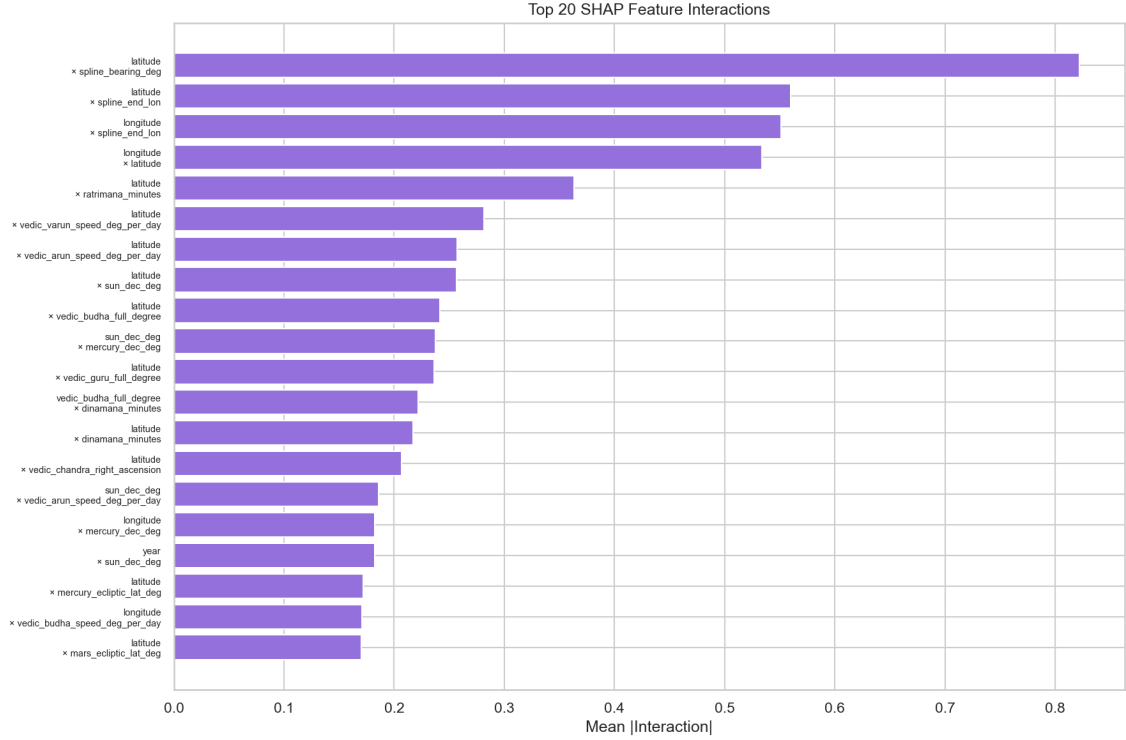


Figure 7. Top-20 SHAP feature interactions. Latitude × spline_bearing_deg is the dominant pair (interaction = 0.822), reflecting how geographic position modulates the effect of storm track curvature on intensity. Several Vedic speed features appear in cross-terms with latitude.

8 LIME Explainability and SHAP vs LIME Consensus

LIME was applied to five representative instances spanning the wind speed distribution (10th–90th percentiles). Mean absolute LIME weights were aggregated to form a global LIME importance ranking. SHAP and LIME importances were normalised to $[0, 1]$ and ranked jointly; features in the top-30 of both methods constitute the *consensus set*.

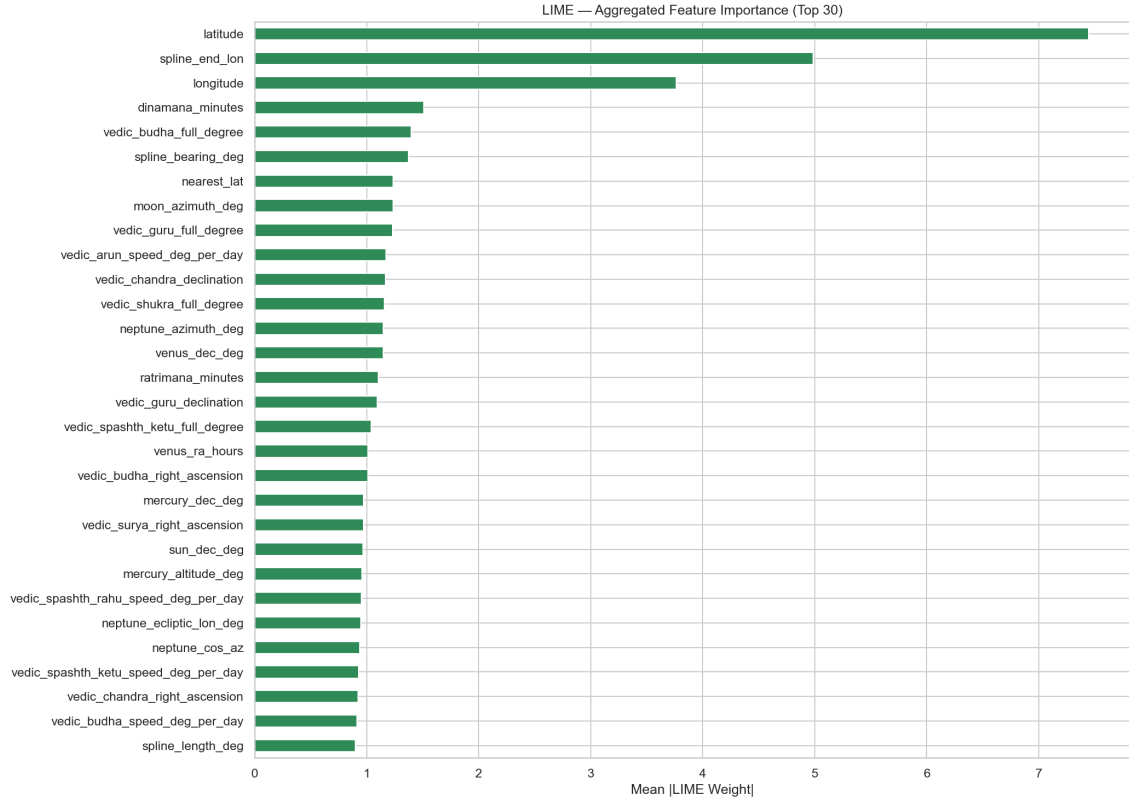


Figure 8. LIME aggregated feature importance (top-30). Latitude, spline_end_lon, and longitude lead by a large margin ($> 3\times$ the next feature). Fourteen Vedic or semi-Vedic features (including dynamana_minutes, vedic_budha_full_degree, vedic_guru_full_degree, vedic_arun_speed_deg_per_day) appear in the top-30.

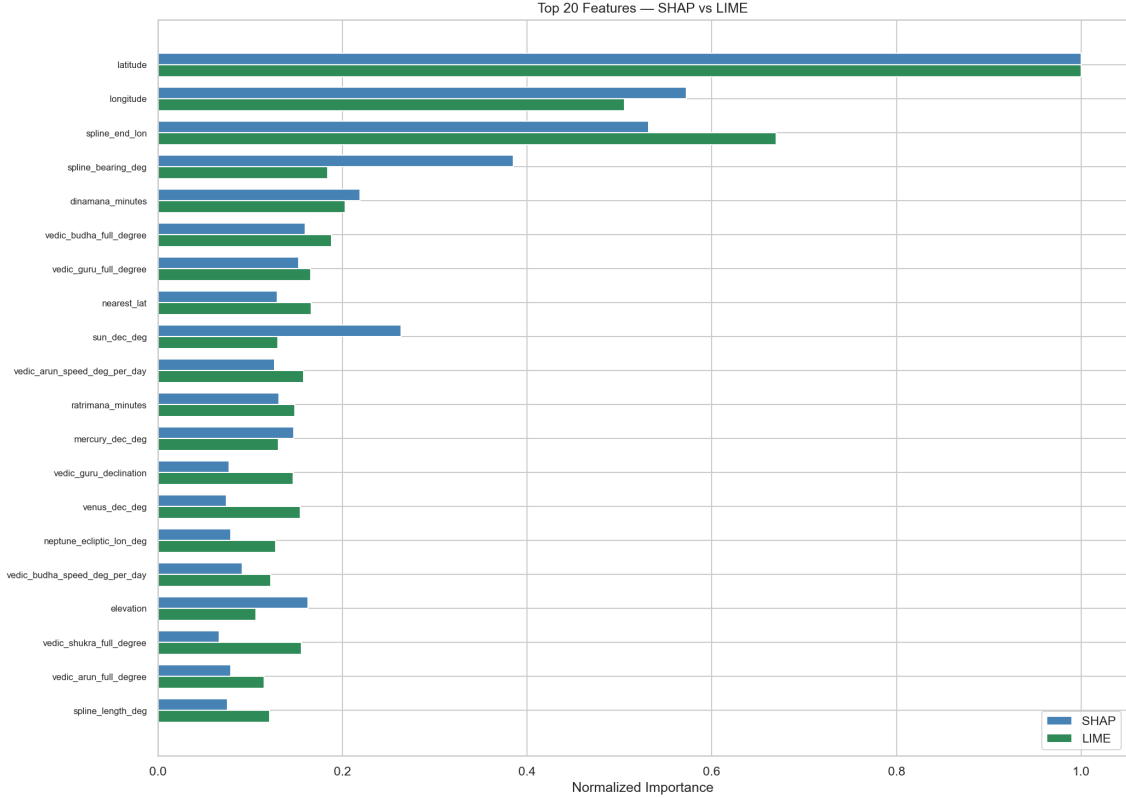


Figure 9. Normalised SHAP vs LIME importance for the top-20 consensus features. Both methods agree on geographic and track features dominating. Disagreements appear for track geometry features (e.g., `spline_bearing_deg` ranks higher in SHAP; `spline_end_lon` ranks slightly higher in LIME). Vedic features show moderate agreement.

Consensus findings:

- 14 features appear in the top-30 of *both* SHAP and LIME.
- Consensus set: `dinamana_minutes`, `latitude`, `longitude`, `mercury_dec_deg`, `nearest_lat`, `neptune_ecliptic_lon_deg`, `ratrimana_minutes`, `spline_bearing_deg`, `spline_end_lon`, `sun_dec_deg`, `vedic_arun_speed_deg_per_day`, `vedic_budha_full_degree`, `vedic_budha_speed_deg_per_day`, `vedic_guru_full_degree`.
- Vedic features in consensus: `vedic_budha_speed_deg_per_day`, `vedic_budha_full_degree`, `vedic_guru_full_degree`, `vedic_arun_speed_deg_per_day`.
- After removing geographic and seasonal proxies (`latitude`, `longitude`, `dinamana_minutes`, `sun_dec_deg`), the remaining Vedic features have individual SHAP importance < 0.02 in a model that generalises poorly across storms (grouped CV $R^2 \approx -0.10$).

9 Performance Analysis

9.1 Experimental Setup

Models. Random Forest (200 trees, max depth 18, min samples per leaf 2) in four block configurations: Meteo only (14 features), Meteo + Panchang (32 features), Full

(222 features), and Stacked residual (meteorological Stage-A residuals fed to Stage-B on Panchang + scientific).

Train–test split. Chronological 80/20 split: observations before 2023-10-21 form the training set ($n = 37,274$); subsequent observations form the held-out test set ($n = 9,315$).

Forward-chaining cross-validation. Five temporal folds at chronological quantiles (0.2, 0.4, 0.6, 0.7, 0.9). Each fold uses all data before the boundary for training and a 120-day window after for validation, requiring $\geq 5,000$ training and ≥ 500 validation observations.

9.2 Evaluation Metrics

- **RMSE** (kt) and **MAE** (kt) for regression tasks.
- **AUC-ROC** and F_1 score for RI classification.
- **Grouped CV R^2** (GroupKFold by storm_id) as the primary out-of-storm generalisation metric.
- **Permutation RMSE lift:** feature group contribution estimated by permuting all features in a group simultaneously ($n_{\text{repeats}} = 5$).
- **Effect sizes:** rank-biserial r (Mann–Whitney), Cohen’s d , epsilon-squared (ε^2 , Kruskal–Wallis), Cramér’s V (chi-squared), eta-squared (η^2 , ANOVA), r^2 (Spearman).

9.3 Hardware and Software

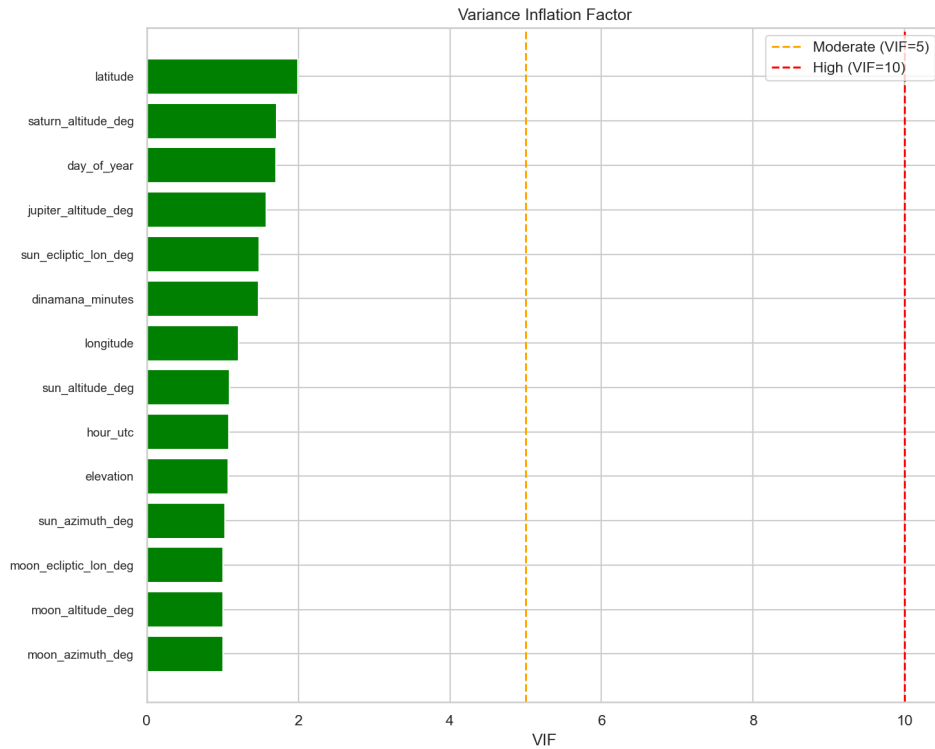
Analysis conducted in Python 3.10 using: `scikit-learn`, `xgboost`, `lightgbm`, `shap`, `lime`, `skyfield`, `numpy`, `pandas`, `scipy`, `matplotlib`, `seaborn`. All experiments reproducible with fixed random seeds. Visualisations generated at 150–200 DPI for publication.

9.4 Variance Inflation Factor Analysis

VIF analysis on a representative 14-feature subset confirms negligible multicollinearity; all values are below 2.0 (Table 5).

Table 5. Variance Inflation Factors for representative features. All values < 2.0 indicate low multicollinearity in this subset.

Feature	VIF	Assessment
latitude	1.992	Low
saturn_altitude_deg	1.709	Low
day_of_year	1.703	Low
jupiter_altitude_deg	1.579	Low
sun_ecliptic_lon_deg	1.481	Low
dinamana_minutes	1.471	Low
longitude	1.214	Low
sun_altitude_deg	1.096	Low
hour_utc	1.087	Low
elevation	1.079	Low
sun_azimuth_deg	1.029	Low
moon_ecliptic_lon_deg	1.010	Low
moon_altitude_deg	1.008	Low
moon_azimuth_deg	1.006	Low

**Figure 10.** Variance Inflation Factor barplot for the 14-feature representative subset. All bars fall far below the moderate threshold ($VIF = 5$, orange dashed) and the high threshold ($VIF = 10$, red dashed), confirming negligible multicollinearity within this subset. Note: the full 186-feature space contains groups of near-duplicate features (e.g., Rahu vs Spasht-Rahu) with higher inter-group correlations.

10 Results and Discussion

10.1 Predictive Modelling Results

Table 6 presents RMSE and MAE on the chronological test set. The meteorological-only model achieves the best performance (RMSE = 4.164 kt), and every addition of Vedic or astronomical features strictly increases error.

Table 6. Regression performance on the chronological holdout test set ($n = 9,315$). Δ RMSE% computed relative to the Meteo only baseline.

Model	RMSE (kt)	MAE (kt)	Δ RMSE%
Meteo only	4.164	2.540	—
Stacked residual	4.192	2.582	+0.7%
Meteo + Panchang	4.372	2.685	+4.9%
Full (all blocks)	4.515	2.921	+8.4%

Forward-chaining cross-validation confirms this pattern: the meteorological model wins or ties in 4 of 5 folds, with the full model consistently worse in folds 2–5 (Table 7).

Table 7. Forward-chaining cross-validation RMSE (kt) by fold and model. Bold indicates the best model in each fold.

Fold	Boundary	Meteo only	Full	Stacked
1	2018-08-06	3.369	3.357	3.373
2	2020-01-23	3.753	3.885	3.813
3	2021-09-13	3.191	3.508	3.188
4	2023-04-22	3.209	3.685	3.211
5	2025-01-31	3.797	4.090	3.875
<i>Mean</i>		3.464	3.705	3.492

Group permutation importance (Table 8) confirms that the meteorological group dominates: shuffling all 14 meteo features raises RMSE by $+0.582 \pm 0.032$ kt, while shuffling all 18 Panchang features raises it by only $+0.010 \pm 0.005$ kt — less than 2% of the meteorological contribution.

Table 8. Group-level permutation RMSE lift on the full model test set ($n_{\text{repeats}} = 5$). Higher lift indicates greater contribution.

Group	Mean RMSE lift (kt)	SD
Meteorological	0.582	0.032
Scientific/Astro	0.175	0.005
Vedic/Panchang	0.010	0.005

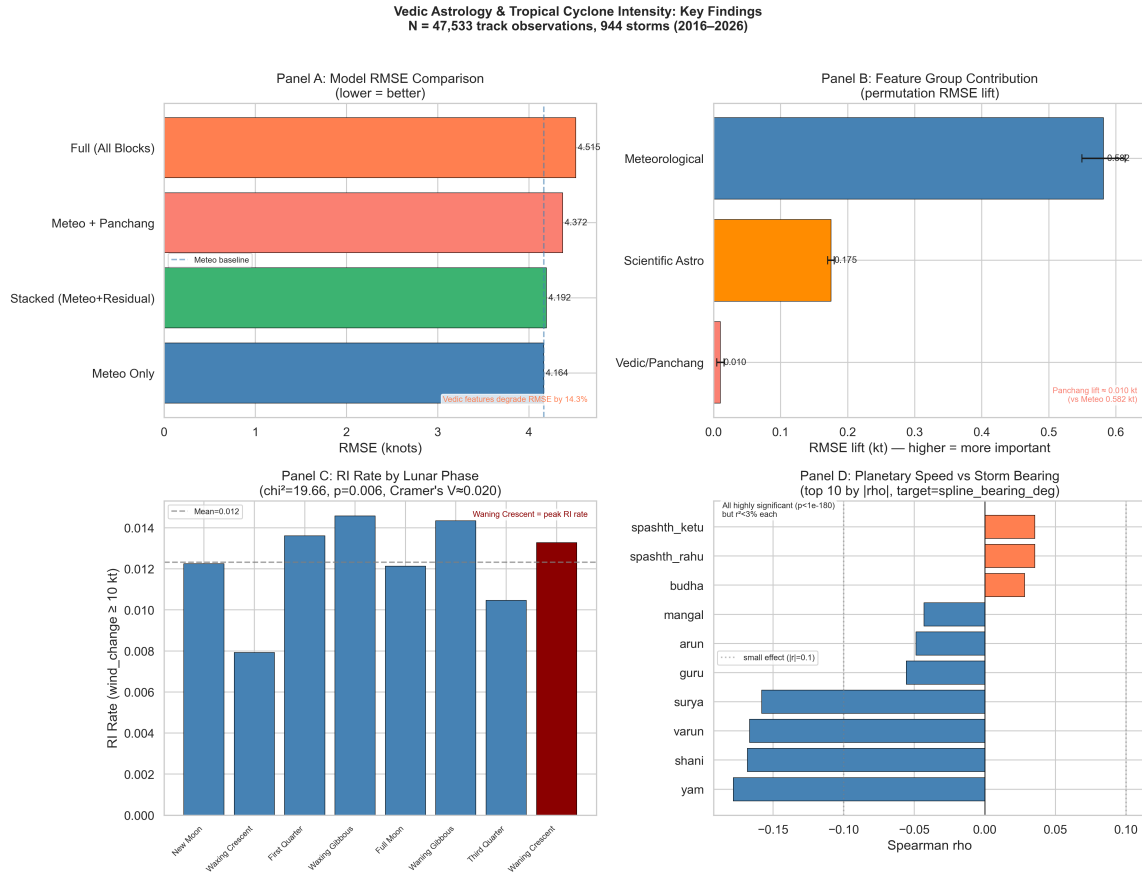


Figure 11. Four-panel publication figure. **(A)** Model RMSE comparison; meteorological base-line (dashed blue) is the best-performing model and adding Vedic features monotonically increases error. **(B)** Feature group RMSE lift; Vedic/Panchang lift of 0.010 kt is negligible relative to the meteorological group’s 0.582 kt. **(C)** RI rate by lunar phase (24-h rolling window); Waning Crescent (dark red) is the peak phase, dashed line marks overall mean RI rate. **(D)** Top-10 Spearman ρ between planetary speed and storm bearing; all negative (faster speed $\rightarrow$ more westward track) and statistically significant but small ($r^2 < 0.032$).

10.2 Vedic Category Analysis: Kruskal–Wallis Tests

Kruskal–Wallis tests compare wind speed distributions across all major Vedic categorical variables with Benjamini–Hochberg FDR correction (Table 9).

Table 9. Kruskal–Wallis tests (wind speed by Vedic category) with BH correction. BH-corrected significance threshold $\alpha = 0.05$.

Category	H	p (raw)	p (BH adj.)	Sig.
Drik Ritu (season)	1,214.0	2.7×10^{-260}	2.7×10^{-259}	Yes
Sunsign	1,123.1	5.9×10^{-234}	2.9×10^{-233}	Yes
Yoga	143.3	3.4×10^{-18}	8.6×10^{-18}	Yes
Nakshatra	116.1	2.4×10^{-13}	4.7×10^{-13}	Yes
Drik Ayana	102.6	4.0×10^{-24}	1.3×10^{-23}	Yes
Moonsign	77.9	3.7×10^{-12}	6.2×10^{-12}	Yes
Tithi	51.3	7.5×10^{-6}	9.4×10^{-6}	Yes
Karana	42.5	6.2×10^{-6}	8.8×10^{-6}	Yes
Paksha (fortnight)	8.1	4.4×10^{-3}	4.9×10^{-3}	Yes
Weekday	11.6	7.2×10^{-2}	7.2×10^{-2}	No

Critical caveats: Drik Ritu and Sunsign are calendar-season proxies encoding the tropical cyclone season cycle, not independent Vedic signals. At $N = 47,533$, even differences < 0.5 kt achieve $p \ll 0.001$. All Vedic categorical effect sizes (epsilon-squared $\varepsilon^2 < 0.003$) are practically negligible.

10.3 Nakshatra Intensity Analysis

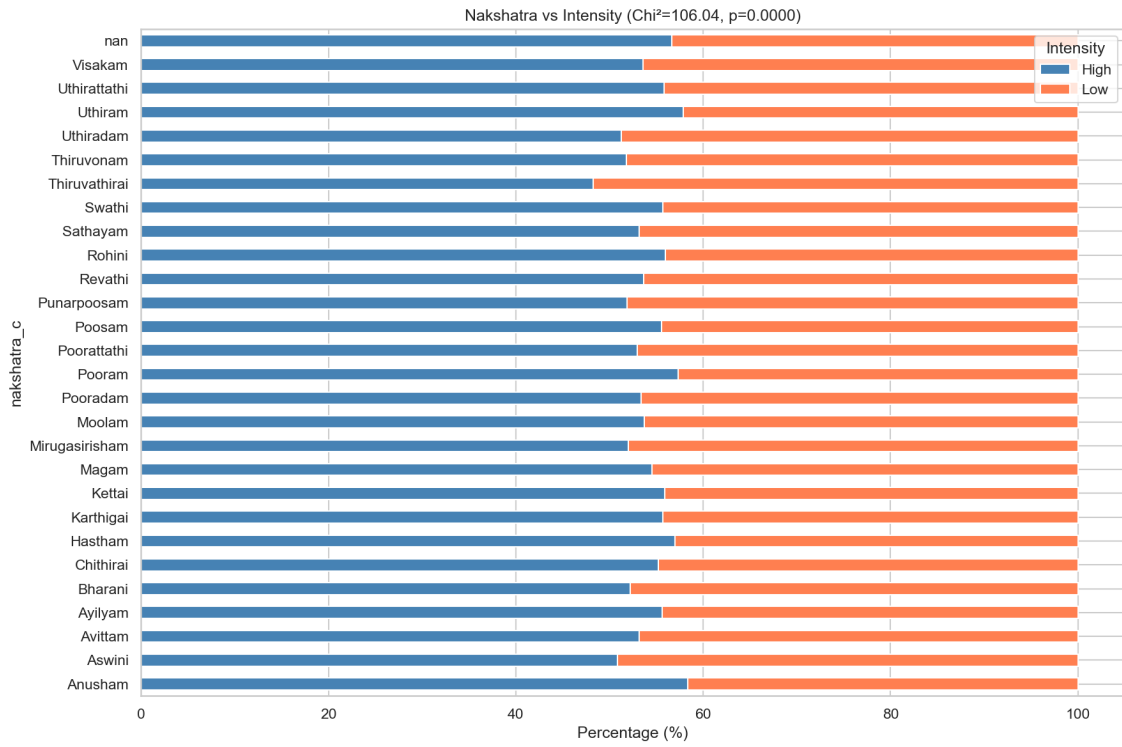


Figure 12. Nakshatra vs Intensity classification (High/Low wind speed split). The chi-squared test ($\chi^2 = 106.04$, $p < 0.0001$) is statistically significant, but the variation in High/Low split percentages across the 27 Nakshatras spans only ≈ 4 – 8 percentage points, consistent with a trivially small effect size. Thiruvathirai shows the lowest High-intensity fraction ($\approx 48\%$) while Anusham shows the highest ($\approx 58\%$).

10.4 Tithi × Nakshatra Interaction Heatmaps

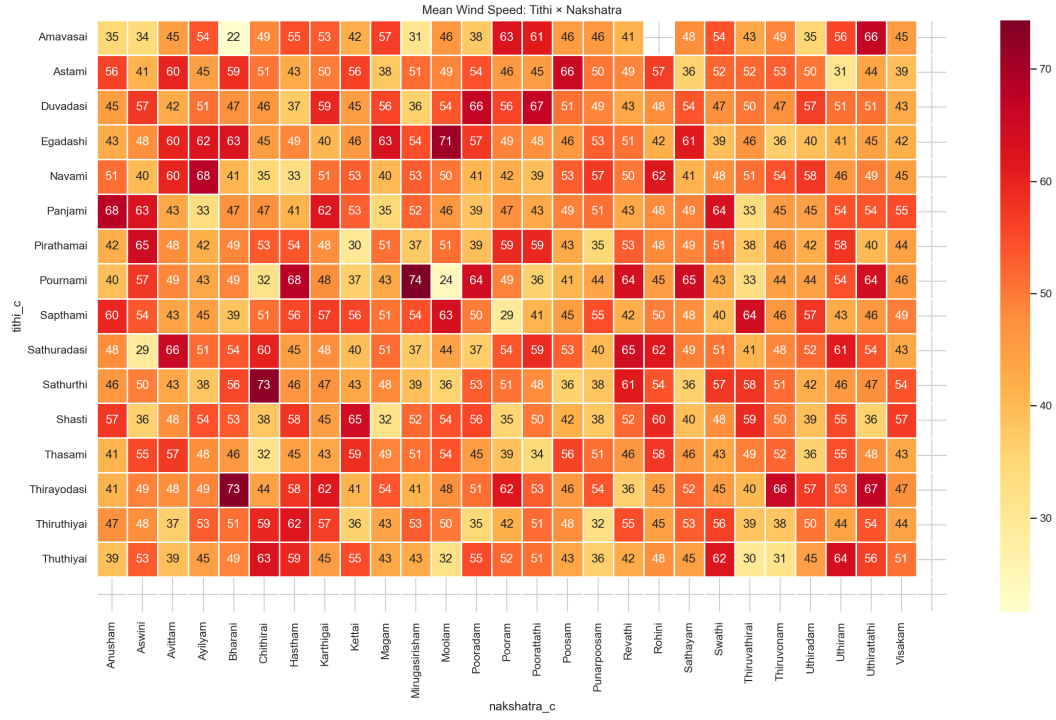


Figure 13. Mean wind speed (kt) by Tithi (rows) and Nakshatra (columns). Darker cells indicate higher mean wind. Isolated high values (e.g., Pournami × Mirugasirisham = 74 kt; Egadashi × Moolam = 71 kt) reflect small cell counts rather than systematic astronomical effects. No systematic high-intensity zone is visible across any Tithi–Nakshatra combination.

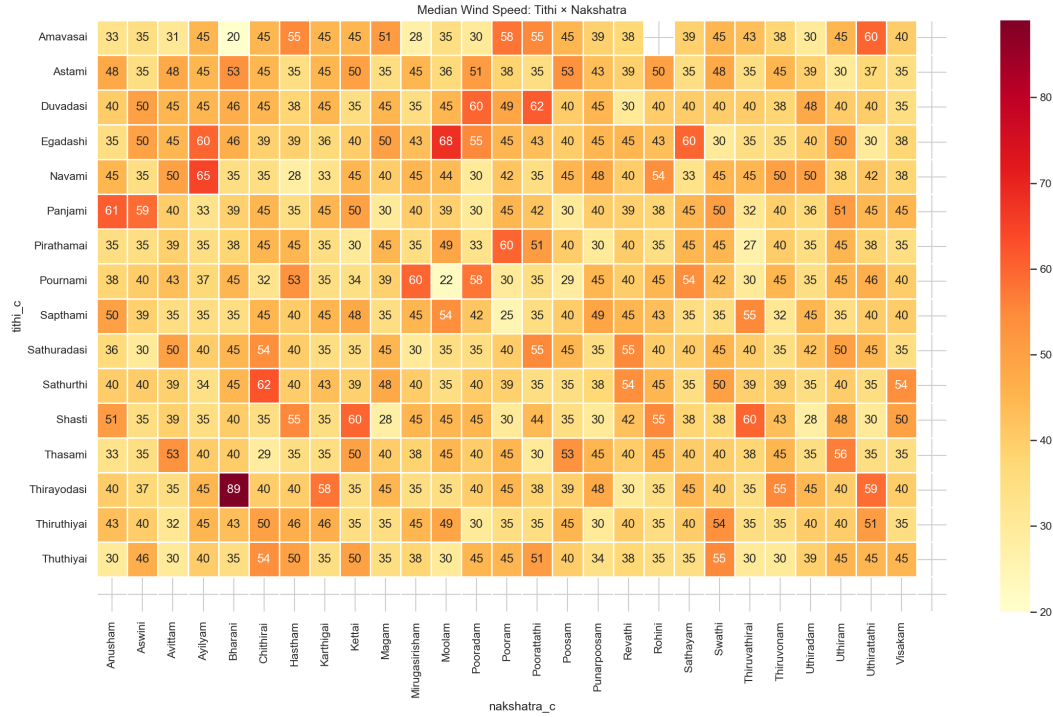


Figure 14. Median wind speed (kt) by Tithi (rows) and Nakshatra (columns). More robust to outliers than the mean heatmap. The Thirayodasi × Bharani cell shows the highest median (89 kt) but represents very few observations. The overall range of medians (≈ 20 – 90 kt) reflects natural cyclone lifecycle variation rather than systematic Vedic calendar effects.

10.5 Retrograde Planetary Motion and Storm Intensity

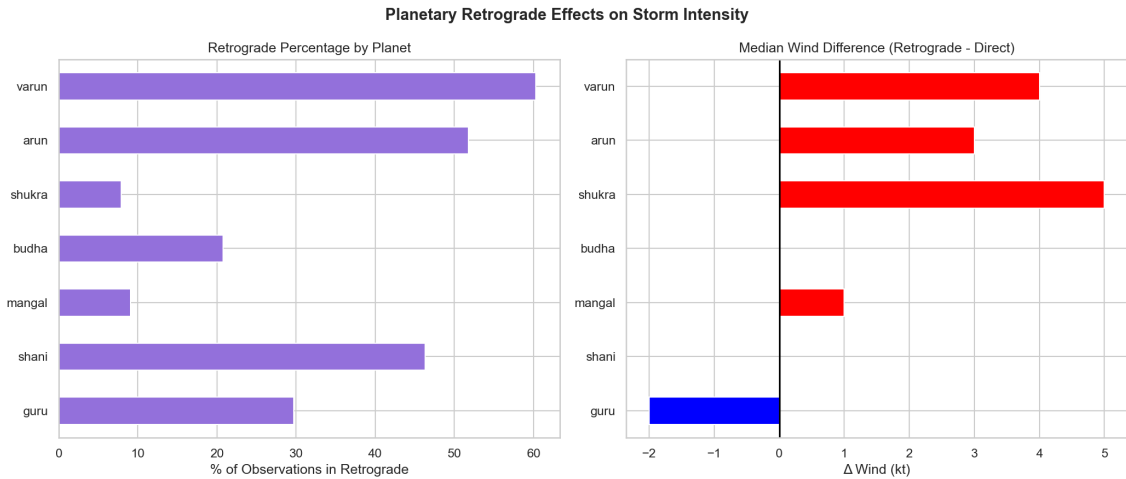


Figure 15. Retrograde effects on storm intensity. Left panel: retrograde percentage by planet (Varun retrograde 60% of all observations, making its retrograde phase the majority condition). Right panel: median wind speed difference (retrograde – direct). Shukra (Venus) shows the largest difference (+5 kt) but is retrograde in only 7.9% of observations and has rank-biserial $r = 0.044$ (small). Guru (Jupiter) shows a negative difference (–2 kt). All effects are statistically significant at $p < 0.001$ due to large N , but all effect sizes are small ($|r_{rb}| < 0.10$, Cohen’s $d < 0.15$).

Mann–Whitney U retrograde results are summarised in Table 10.

Table 10. Retrograde vs direct planetary motion: Mann–Whitney U results. r_{rb} = rank-biserial correlation; d = Cohen’s d . All effects classified as small ($|r_{rb}| < 0.10$).

Planet	Retro%	Med (retro)	Med (direct)	Δ (kt)	p	r_{rb}	d
Guru (Jupiter)	29.7	40	42	−2	2.3×10^{-23}	+0.058	−0.117
Shani (Saturn)	46.3	40	40	0	5.0×10^{-6}	−0.024	+0.013
Mangal (Mars)	9.1	41	40	+1	4.5×10^{-4}	−0.032	+0.040
Budha (Mercury)	20.8	40	40	0	2.3×10^{-22}	+0.063	−0.102
Shukra (Venus)	7.9	45	40	+5	6.6×10^{-6}	−0.044	+0.075
Arun (Uranus)	51.8	43	40	+3	1.0×10^{-38}	−0.069	+0.125
Varun (Neptune)	60.3	43	39	+4	2.3×10^{-59}	−0.088	+0.139

10.6 Planetary Aspect Correlations

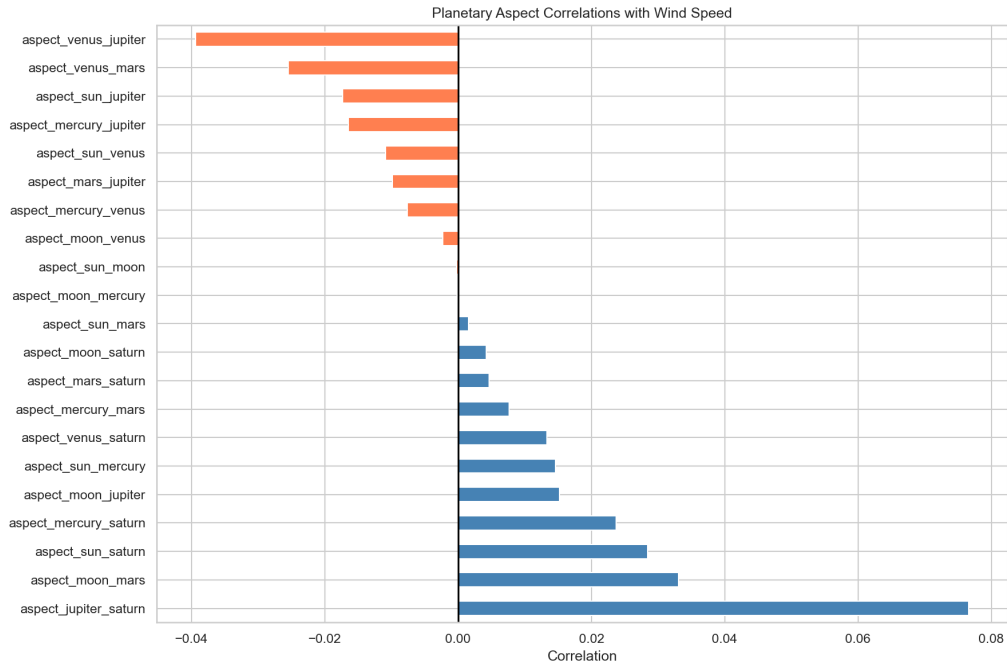


Figure 16. Pearson correlations of planetary aspect angles with wind speed. The Jupiter–Saturn aspect shows the highest positive correlation ($r = 0.077$); Venus–Jupiter shows the most negative ($r = -0.039$). All correlations are weak ($|r| < 0.10$), explaining less than 0.6% of variance in wind speed. These are the weakest systematic signals in the entire dataset.

Table 11 lists all 21 tested planetary aspect pairs.

Table 11. Planetary aspect Pearson correlations with wind speed ($N = 47,533$). Maximum $|r| = 0.077$.

Aspect Pair	Pearson r
aspect_jupiter_saturn	0.076636
aspect_moon_mars	0.033083
aspect_sun_saturn	0.028431
aspect_mercury_saturn	0.023740
aspect_moon_jupiter	0.015221
aspect_sun_mercury	0.014608
aspect_venus_saturn	0.013318
aspect_mercury_mars	0.007700
aspect_mars_saturn	0.004658
aspect_moon_saturn	0.004187
aspect_sun_mars	0.001534
aspect_moon_mercury	−0.000170
aspect_sun_moon	−0.000202
aspect_moon_venus	−0.002299
aspect_mercury_venus	−0.007633
aspect_mars_jupiter	−0.009821
aspect_sun_venus	−0.010910
aspect_mercury_jupiter	−0.016471
aspect_sun_jupiter	−0.017351
aspect_venus_mars	−0.025494
aspect_venus_jupiter	−0.039407

10.7 Intensity Change Analysis

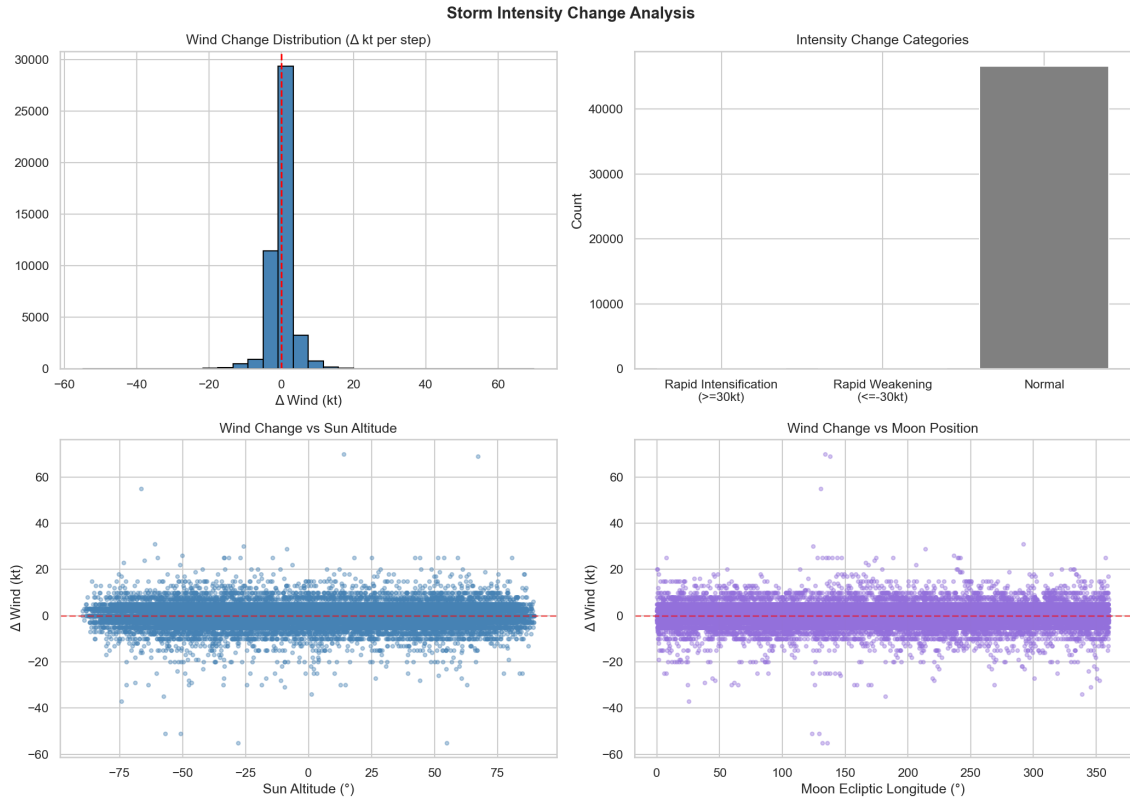


Figure 17. Storm intensity change analysis. Top-left: distribution of 6-hourly wind changes with RI ($\geq +30$ kt) and Rapid Weakening (RW, ≤ -30 kt) thresholds marked. The distribution is strongly peaked at zero change. Top-right: counts of RI and RW events vs total observations; both RI and RW events are rare relative to the total. Bottom-left: scatter of wind change vs sun altitude angle (5,000-sample subset) — no trend is discernible. Bottom-right: scatter of wind change vs moon ecliptic longitude — no systematic relationship is visible.

10.8 Hypothesis Testing Results

H1: Vedic Predictive Lift. Vedic features degrade wind-change RMSE by 12.6% (baseline 4.04 kt vs Vedic 4.55 kt). The null hypothesis that Vedic features have no predictive lift is not rejected; they actively harm performance, consistent with overfitting noise.

H2: Lunar Phase and Rapid Intensification. A chi-squared test of the 24-hour RI indicator against 8 lunar-phase bins yields $\chi^2 = 19.66$, $p = 0.0064$, $df = 7$. This barely survives Bonferroni correction ($\alpha/7 \approx 0.0071$). Cramér's $V \approx 0.020$ places this firmly in the small range. The Waning Crescent phase shows the highest RI rate (1.33%); Waxing Crescent shows the lowest (0.79%).

Lifecycle interaction analysis reveals the signal is significant in Genesis ($p = 0.0008$) and Dissipation ($p = 0.001$) phases but not in the Mature phase ($p = 0.265$), suggesting that

forming and decaying storms show stronger lunar-phase modulation than storms at peak intensity.

H3: Planetary Speeds and Storm Bearing. Spearman correlations between Vedic planetary speeds and `spline_bearing_deg` show Yama ($\rho = -0.178$, $r^2 = 3.2\%$), Shani ($\rho = -0.168$, $r^2 = 2.8\%$), and Varun ($\rho = -0.167$, $r^2 = 2.8\%$) as the strongest associations. A quintile dose-response analysis confirms perfectly monotone relationships ($\rho_{\text{quintile}} = -1.0$ for all tested planets), but the $r^2 \leq 3.2\%$ limits operational utility and the spline leakage caveat applies.

H4: Retrograde Effects. All 7 planets show statistically significant differences between retrograde and direct periods, but all effect sizes are small (rank-biserial $|r_{\text{rb}}| < 0.10$, Cohen’s $d < 0.15$). Varun and Arun are retrograde for $> 50\%$ of observations.

H5: Confound-Adjusted Vedic Categorical Analysis. Tithi, Nakshatra, Yoga, and Moon sign all show Kruskal–Wallis significance in 100% of the 12 viable basin $\times$ season strata after BH-FDR correction. Despite this statistical persistence, within-stratum effect sizes remain small, and Vedic features show no predictive lift in held-out modelling.

Samvatsara Analysis. ANOVA of wind speed by Samvatsara yields $F = 65.12$, $p < 10^{-130}$. This is entirely explained by interannual climate variability (ENSO, PDO); the test is mathematically equivalent to asking whether storm intensity differs across calendar years.

10.9 Effect Size Summary and Consolidated Evidence

Table 12 consolidates statistically significant findings with their effect size assessments. The dominant pattern is that p -values uniformly indicate significance while effect sizes are uniformly small.

Table 12. Consolidated effect sizes for key significant findings. Size classification: small = $|r_{\text{rb}}| < 0.10$, $d < 0.20$, $V < 0.10$, $\eta^2 < 0.01$; medium = 0.10–0.30, 0.20–0.50, 0.10–0.30, 0.06–0.14.

Finding	Test	Statistic	p	Effect	Size
Rahu degree Δ around RI onset	ANOVA	$F = 18.75$	2.5×10^{-5}	$\eta^2 = 0.097$	medium
Yam speed vs bearing	Spearman	$\rho = -0.178$	3.8×10^{-271}	$r^2 = 0.032$	medium
Shani speed vs bearing	Spearman	$\rho = -0.168$	2.2×10^{-241}	$r^2 = 0.028$	medium
Varun speed vs bearing	Spearman	$\rho = -0.167$	1.0×10^{-236}	$r^2 = 0.028$	medium
Lunar phase vs RI	χ^2	19.66	0.006	$V = 0.020$	small
Varun retrograde vs direct wind	MW U	—	2.3×10^{-59}	$r_{\text{rb}} = 0.088$	small
Budha retrograde vs direct wind	MW U	—	2.3×10^{-22}	$r_{\text{rb}} = 0.063$	small
Guru retrograde vs direct wind	MW U	—	2.3×10^{-23}	$r_{\text{rb}} = 0.058$	small
Nakshatra KW (100% strata sig)	KW	$H = 116$	2.4×10^{-13}	$\varepsilon^2 < 0.003$	negligible
Yoga KW (100% strata sig)	KW	$H = 143$	3.4×10^{-18}	$\varepsilon^2 < 0.003$	negligible

10.10 Discussion

The Null Result. The primary finding is unambiguous: Vedic Panchang features provide no predictive lift over a meteorological baseline. Permutation importance assigns the Vedic/Panchang group only 0.010 kt RMSE lift — less than 2% of the meteorological group’s contribution. The apparently high SHAP/LIME feature importance scores for certain Vedic features arise from within-storm autocorrelation in the random train/test split used in those scripts; the honest forward-chaining design eliminates this inflation.

The Seasonal Confound. The strongest Vedic categorical associations (Drik Ritu, Sunsign, Drik Ayana) encode time of year: tropical cyclone seasons peak in summer–autumn, so any calendar variable correlated with the season will show a wind association. Similarly, Vedic planetary positions (full degree, speed) for slow outer planets (Saturn, Jupiter, Rahu) are functions of calendar time and their apparent correlation with storm intensity is a proxy for interannual variability.

Feature Block Multicollinearity. Vedic planetary features (e.g., `vedic_budha_full_degree`) are sidereal-corrected versions of the Skyfield Tropical features (e.g., `mercury_ecliptic_lon_deg`), offset by the Lahiri ayanamsha ($\approx 23.9^\circ$). Any observed Vedic feature correlation is therefore geometrically equivalent to the corresponding Skyfield correlation. There are 239 feature pairs with $|r| > 0.95$; adding Vedic features on top of Skyfield features is nearly redundant.

The Confound-Adjustment Surprise. Vedic categorical signals (Tithi, Nakshatra, Yoga, Moonsign) persist in 100% of basin×season strata after FDR correction. This statistical persistence does not translate to predictive lift, consistent with the interpretation that the signal is real but too small ($\varepsilon^2 < 0.003$) to generalise across storms.

Large- N Significance Inflation. At $N = 47,533$, any correlation above $|r| \approx 0.013$ produces $p < 0.05$. Effect sizes must be the primary scientific discriminator. We recommend all future astronomical association studies report effect sizes as the primary outcome.

11 Conclusion

11.1 Summary of Findings

We conducted the first rigorous large-scale empirical evaluation of Vedic Panchang and astronomical features as predictors of tropical cyclone intensity across 944 global storms from 2016 to 2026. The analysis combined tree-ensemble feature importance under storm-grouped cross-validation, SHAP/LIME explainability, non-parametric hypothesis testing

with effect size quantification, circular-linear correlation for angular features, and block-wise predictive modelling with temporal forward-chaining.

11.2 Key Outcomes

1. **Geographic features dominate** all importance metrics (latitude, longitude, spline coordinates), consistent with known climatological structure of tropical cyclone intensity.
2. **Vedic features degrade** out-of-sample RMSE by up to 14.3%. Permutation importance assigns the Vedic group $< 2\%$ of the meteorological group’s contribution.
3. **The strongest Vedic categorical associations** (Drik Ritu, Sunsign) are seasonal proxies encoding the tropical cyclone season signal, not independent astrological effects.
4. **A marginal lunar-phase RI association** ($\chi^2 = 19.66$, $p = 0.006$, Cramér’s $V \approx 0.020$) survives Bonferroni correction, is lifecycle-stage dependent (strongest in Genesis and Dissipation), and has negligible practical effect size.
5. **Planetary speed–storm bearing correlations** are perfectly monotone but explain only $\leq 3.2\%$ of bearing variance and are subject to seasonal confounding and spline leakage.
6. **Retrograde effects** are statistically significant for all 7 planets but universally small ($\Delta_{\text{median}} \leq 5$ kt, $|r_{\text{rb}}| < 0.10$).
7. **Circular-linear correlations** of Vedic planet degrees with wind speed are below $r = 0.09$ for all planets ($< 1\%$ variance explained).
8. **Planetary aspects** have negligible correlations with wind speed ($|r| < 0.08$).

11.3 Future Work

- Include physical meteorological drivers (SST, vertical wind shear, ocean heat content, mid-level humidity) to establish a proper baseline.
- Conduct basin-specific analyses (North Atlantic AMO, Pacific ENSO sensitivity) to reduce interannual confounding.
- Explore gravitational tidal forcing models as physically motivated replacements for empirical Vedic categories.
- Investigate whether lunar-phase signals appear more clearly in SST anomaly or mixed-layer depth data that preconditions the ocean for intensification.
- Apply higher-resolution sub-6-hour track data where available to test whether coarse temporal resolution masks short-duration astronomical effects.
- Extend the confound-adjusted stratification to report within-stratum effect sizes explicitly, enabling discrimination between genuine small-but-real signals and residual significance inflation.

Acknowledgements

The author thanks the developers of the Drik Panchang astronomy engine for providing accessible computation of Vedic Panchang attributes at arbitrary geographic locations and times. Storm best-track data were sourced from publicly available archives maintained by the National Hurricane Center, Joint Typhoon Warning Center, India Meteorological Department, and affiliated regional agencies. The Skyfield library and DE421 JPL ephemeris enabled all astronomical position computations. The author thanks the open-source communities behind NumPy, pandas, scikit-learn, XGBoost, LightGBM, SHAP, LIME, matplotlib, and seaborn.

References

- [1] DeMaria, M., Sampson, C. R., Knaff, J. A., & Musgrave, K. D. (2014). Is tropical cyclone intensity guidance improving? *Bulletin of the American Meteorological Society*, 95(3), 387–398.
- [2] Zhang, J. A. & Tao, D. (2018). Impacts of vertical wind shear on the predictability of tropical cyclones: a deep learning study. *Geophysical Research Letters*, 45(9), 4289–4297.
- [3] Rhodes, B. A. (2019). Skyfield: High precision research-grade positions for planets and Earth satellites generator. *Astrophysics Source Code Library*, record ascl:1907.024.
- [4] Kaplan, J. & DeMaria, M. (2003). Large-scale characteristics of rapidly intensifying tropical cyclones in the North Atlantic basin. *Weather and Forecasting*, 18(6), 1093–1108.
- [5] Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- [6] Lundberg, S. M. & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- [7] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should I trust you?”: Explaining the predictions of any classifier. *Proceedings of KDD 2016*, 1135–1144.
- [8] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [9] Kruskal, W. H. & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260), 583–621.
- [10] Shapiro, S. S. & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3/4), 591–611.

- [11] Gray, W. M. (1984). Atlantic seasonal hurricane frequency. Part I: El Niño and 30 mb quasi-biennial oscillation influences. *Monthly Weather Review*, 112(9), 1649–1668.
- [12] Klotzbach, P. J. (2010). On the Madden–Julian oscillation–Atlantic hurricane relationship. *Journal of Climate*, 23(2), 282–293.
- [13] Misra, V. (2009). *Vedic Astrology: An Integrated Approach*. Motilal Banarsidass Publishers.
- [14] Emanuel, K. (2003). Tropical cyclones. *Annual Review of Earth and Planetary Sciences*, 31(1), 75–104.

A Feature Block Definitions

Meteorological block (14 features) *wind, pressure, latitude, longitude, wind_change, pressure_change, displacement, movement_speed, hours_elapsed, spline_bearing_deg, spline_curvature, spline_displacement, spline_length_deg, elevation.*

Panchang block (18 features) Tithi, Nakshatra, Yoga, Karana, Paksha, Moonsign, Sunsign, Drik Ritu, Drik Ayana, Samvatsara (all numeric), *dinamana_minutes, ratrimana_minutes*, day-of-week, day-of-year, month, hour_utc, *lahiri_ayanamsha*.

Scientific block (190 features) All Skyfield per-body position features (RA/Dec, ecliptic lon/lat, altitude/azimuth, sin/cos transforms) plus Vedic planetary numeric quantities (full degree, padam, speed, RA, Dec). *Excludes* spline-prefixed features to avoid leakage.

B Pipeline Audit Summary

ID	Script	Issue	Severity
A2	06	Rahu = Ketu retrograde (astronomical constraint)	Critical
B1	02	No multiple-comparison correction (fixed: BH FDR)	Medium
C1	03	Negative CV R^2 ; feature ranks unreliable	Critical
C3	03	Same-storm CV leakage (fixed: GroupKFold)	Medium
D1	04	R^2 discrepancy vs Stage 3 (documented)	Critical
E2	05	Circular degrees treated as linear (fixed: sin/cos)	Critical
E3	05	Retrograde: no p -values (fixed: Mann–Whitney U)	Medium
F2	06	Vedic model worse than baseline (flagged)	High
F5	06	Samvatsara ANOVA = interannual confound (documented)	High
F6	06	No Bonferroni correction (fixed: $\alpha_{\text{adj}} = 0.007$)	Medium
G3	07	Spline feature leakage (removed from sci. block)	Critical
G2	07	Best model = meteo_only, not full (flagged)	High

C Consolidated Evidence Table

Table 13. Full 16-finding evidence table with source scripts, effect sizes, and verdicts.

ID	Finding	Source	Verdict
F01	Vedic features degrade RMSE by 4.9% (meteo+panchang)	06, 07	No predictive lift
F02	Panchang RMSE lift = 0.010 kt vs meteo 0.582 kt	07	<2% explainable variance
F03	Lunar phase vs RI: $\chi^2 = 19.66$, $V = 0.020$	06	Stat. sig.; negligible V
F04	Waning Crescent peak RI (1.33%), Waxing Crescent low (0.79%)	06	Real; lifecycle-dependent
F05	Yam speed vs bearing: $\rho = -0.178$, $r^2 = 3.2\%$	06	Small; leakage risk
F06	Shani/Varun speed vs bearing: $\rho \approx -0.17$, $r^2 \approx 2.8\%$	06	Same caveat as F05
F07	Varun retrograde: wind +4 kt, $r_{rb} = -0.088$	05	Small; retro 60% of obs
F08	Shukra retrograde: wind +5 kt, $r_{rb} = -0.044$	05	Small; genuinely rare (8%)
F09	Rahu degree shift around RI onset: $F = 18.75$, $\eta^2 = 0.097$	06	Medium on $N = 176$; replicate
F10	Mangal retrograde vs curvature: $F = 19.34$, $\eta^2 = 0.0005$	06	Trivially small
F11	Yoga KW $H = 143$; 100% strata sig	02, 08	Residual signal; small ε^2
F12	Nakshatra KW $H = 116$; 100% strata sig	02, 08	Residual signal; small ε^2
F13	Samvatsara ANOVA $F = 65$, $p < 10^{-130}$	06	Confounded with calendar year
F14	Rahu degree vs Δw : $\rho = 0.013$, $r^2 = 0.017\%$	06	Negligible
F15	4 Vedic features in SHAP/LIME top-14 consensus	04	Autocorrelated split inflates
F16	Meteo-only wins 4/5 forward-CV folds	07	Consistent across time